

Author Guidelines for the British Machine Vision Conference

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Abstract

Automated baggage screening in security-critical environments faces persistent challenges including extreme class imbalance, limited availability of labeled anomaly samples, and the inherent difficulty of detecting small, heavily occluded threat objects within cluttered X-ray imagery. While supervised deep learning approaches achieve strong detection accuracy, they depend on extensive annotated datasets and frequently fail to generalize to novel or unseen threat types. We propose a **Two-Stage Reconstruction-Guided Hybrid Framework** that combines unsupervised representation learning with lightweight supervised classification for both anomaly detection and spatial localization. In the first stage, a Convolutional Autoencoder (CAE) is trained exclusively on normal baggage images; the resulting pixel-wise reconstruction error, the disparity map serves as the primary anomaly signal, from which discriminative features are extracted to train a Random Forest classifier for image-level detection. In the second stage, a patch-wise reconstruction strategy enables fine-grained spatial analysis, with connected component analysis applied to the thresholded disparity mask to identify and refine anomalous regions as padded bounding boxes. Evaluated on the large-scale SIXray benchmark, the proposed framework achieves a Detection ROC-AUC of 0.916, a detection accuracy of 99.4%, and a localization accuracy of 0.908 at $\text{IoU} \geq 0.30$, outperforming GANomaly, Skip-GANomaly, and Dense-GANomaly baselines while remaining computationally efficient for real-world deployment.

1 Introduction

The security of modern transportation infrastructure depends on the reliable detection of prohibited items within passenger baggage. X-ray screening remains the primary non-intrusive mechanism for this task, enabling inspection of opaque containers without physical search [8]. However, the scale of modern transit operations has made purely manual interpretation untenable: security operators must analyze thousands of cluttered, noisy images per shift under sustained cognitive load, and their attention degrades accordingly [9]. Transitioning to Automated Threat Recognition (ATR) systems has therefore become an operational imperative.

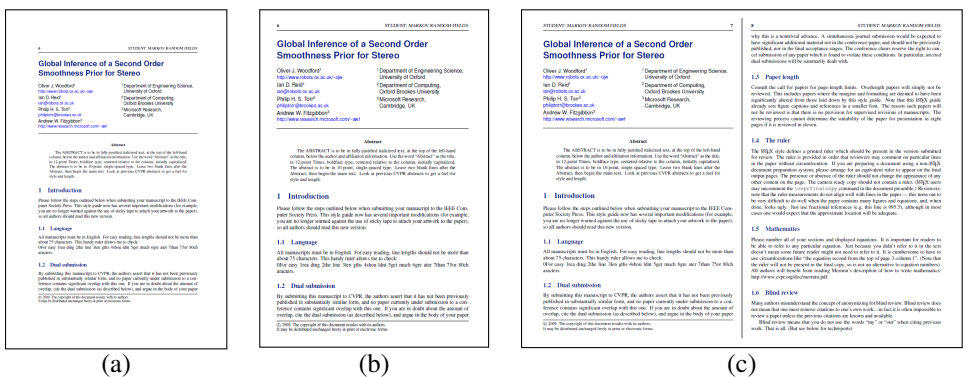
The dominant paradigm in the literature applies supervised deep learning to this problem, adapting established object detection architectures such as YOLO and Faster R-CNN to the X-ray domain [1]. These approaches have demonstrated high detection rates for specific known threat categories. However, they are fundamentally constrained by two interlocking problems. First, the class imbalance in operational environments is extreme: prohibited items appear in a vanishingly small fraction of scans [1], causing supervised models to bias heavily towards the majority (benign) class and produce unacceptably high false-negative rates on rare threats. Second, supervised models operate under a *closed-set* assumption, meaning they can only detect threat types present in their training data. Novel improvised devices, non-metallic 3D-printed firearms, liquid explosives, or concealed components fall outside predefined categories and are effectively invisible to such systems [1]. Acquiring large, balanced, and annotated datasets of real prohibited items is additionally expensive and logistically dangerous.

Unsupervised Anomaly Detection (UAD) offers a compelling alternative by reframing the problem: rather than recognizing specific threats, the system learns what *normal* baggage looks like and flags statistical deviations for human review. Reconstruction-based methods, notably GANomaly [2] and Skip-GANomaly [3], exploit the intuition that an autoencoder trained exclusively on normal images will fail to faithfully reconstruct anomalous content, producing an elevated reconstruction error at the location of the threat. While this approach is theoretically well-motivated, existing adversarial reconstruction methods carry significant computational overhead and, critically, none provide a principled mechanism for precise spatial localization of detected anomalies within the image, a capability that is essential for guiding operator attention in practice.

In this paper, we address these limitations with a **Two-Stage Reconstruction-Guided Hybrid Framework** designed for both accurate image-level detection and fine-grained anomaly localization on X-ray security imagery. Our contributions are as follows:

1. **Reconstruction-Driven Feature Extraction for Classification.** We train a Convolutional Autoencoder solely on normal baggage images and derive a pixel-wise disparity map as the primary anomaly signal. Statistical features extracted from this map are used to train a Random Forest classifier, combining the generalization of unsupervised representation learning with the discriminative power of a supervised classifier, without requiring annotated anomaly samples at training time.
2. **Patch-Wise Localization via Connected Component Analysis.** When an anomaly is detected at the image level, a patch-wise reconstruction strategy applies the autoencoder at fine spatial granularity. A threshold is applied to the resulting disparity mask to produce a binary anomaly map, from which connected components are extracted, size-filtered, and padded to form bounding box predictions.
3. **Comprehensive Evaluation on SIXray.** We benchmark against GANomaly, Skip-GANomaly, and Dense-GANomaly on the SIXray dataset [1], demonstrating that our framework achieves superior detection accuracy (99.4%) and ROC-AUC (0.916), while additionally providing localization performance ($\text{IoU} \geq 0.30$ accuracy of 0.908) that prior baselines do not report.

The remainder of this paper is organized as follows. Section ?? reviews related work in classical, supervised, and unsupervised X-ray threat detection. Section ?? details the proposed two-stage architecture and its loss formulation. Section ?? describes the experimental



(a)

(b)

(c)

Figure 1: It is often a good idea for the first figure to attempt to encapsulate the article, complementing the abstract. This figure illustrates the various print and on-screen layouts for which this paper format has been optimised: (a) traditional BMVC print format; (b) on-screen single-column format, or large-print paper; (c) full-screen two column, or 2-up printing.

setup, datasets, and evaluation metrics. Section ?? presents quantitative and qualitative results, and Section ?? concludes with directions for future work.

1.1 Paper length

Papers must not exceed 14 pages (excluding references). References and any appendices are unlimited. **Only** the acknowledgement and bibliography should be *excluded* from the page count: Papers must not have altered margins and formatting from those laid down by this style guide.

The bibliography should begin immediately after the paper text. It may be of any length, within reason. It must *not* include annotations, figures, or any other paraphernalia intended to subvert the paper length requirement.

1.2 Citations

When citing a multi-author paper, you may save space by using “*et alia*”, shortened to “*et al.*” (not “*et. al.*” as “*et*” is a complete word.) The provided `\etal` macro is a useful *aide memoire* in this regard. However, use it only when there are three or more authors. Thus, the following is correct: “Frobination has been trendy lately. It was introduced by Alpher [9], and subsequently developed by Alpher and Fotheringham-Smythe [10], and Alpher *et al.* [9].”

This is incorrect: “... subsequently developed by Alpher *et al.* [9] ...” because reference [9] has just two authors. If you use the `\etal` macro, then you need not worry about double periods when used at the end of a sentence as in Alpher *et al.*

We use `natbib`, so citations in random order are nicely sorted: [9, 10, 9, 10]. However, we don’t use the `compress` option, as we want each reference to have its own hyperlink and popup window.

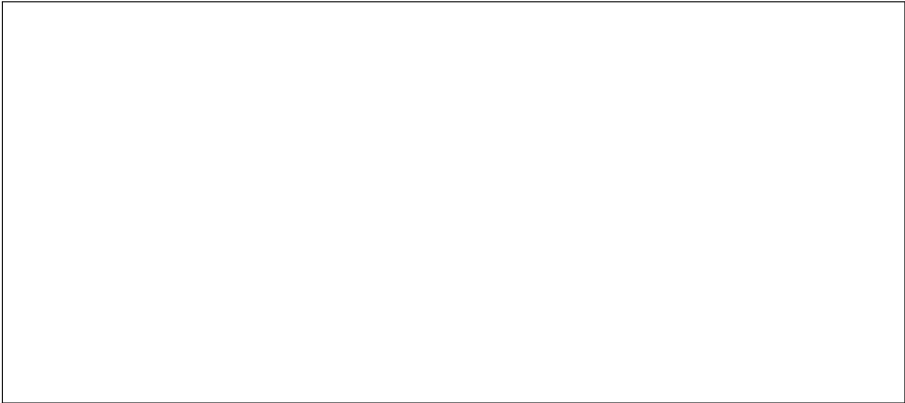


Figure 2: Example of a short caption, which should be centered.

Method	Frobnability
Theirs	Frumpy
Yours	Frobbly
Ours	Makes one’s heart Frob

Table 1: Results. Ours is better.

1.3 Footnotes

Please use footnotes¹ sparingly. Indeed, try to avoid footnotes altogether and include necessary peripheral observations in the text (within parentheses, if you prefer, as in this sentence). If you wish to use a footnote, place it at the bottom of the column on the page on which it is referenced. Use Times 8-point type, single-spaced.

1.4 Mathematics

Please number all of your sections and displayed equations. It is important for readers to be able to refer to any particular equation. Just because you didn’t refer to it in the text doesn’t mean some future reader might not need to refer to it. It is cumbersome to have to use circumlocutions like “the equation second from the top of page 3 column 1”. (Note that the ruler will not be present in the final copy, so is not an alternative to equation numbers). All authors will benefit from reading Mermin’s description [10] of how to write mathematics.

1.5 References

List and number all bibliographical references in 9-point Times, single-spaced, at the end of your paper. When referenced in the text, enclose the citation number in square brackets, for example [1]. Where appropriate, include the name(s) of editors of referenced books.

¹This is what a footnote looks like. It often distracts the reader from the main flow of the argument.

1.6 Color

Color is valuable, and will be visible to readers of the electronic copy. However ensure that, when printed on a monochrome printer, no important information is lost by the conversion to grayscale.

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